INCH-POUND

MIL-DTL-23971D
7 May 2012
SUPERSEDING
MIL-DTL-23971C
20 August 2001

DETAIL SPECIFICATION

POWER DIVIDERS, POWER COMBINERS, AND POWER DIVIDER/COMBINERS,

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for power dividers, power combiners, and power divider/combiners. Specific requirements tailored for “space flight application” components are included to represent the highest reliability components specified herein. Such components are intended for use in NASA manned, satellite, interplanetary or launch vehicle programs.

1.2 Part or Identifying number (PIN). The military PIN consists of the letter "M", the basic number of the specification sheet and an assigned dash number (see 3.1).

23791/1-	01	T
Military designator and specification sheet number	Dash number	Space flight application

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, OH 43218-3990, or emailed to Tubesamps@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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FEDERAL STANDARDS

FED-STD-H28	-	Screw Thread Standards for Federal Services.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-85	-	Waveguides, Rigid, Rectangular, General Specification for.
MIL-DTL-3922	-	Flanges, Waveguide, General Purpose, General Specification for.
MIL-DTL-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-DTL-14072	-	Finishes for Ground Based Electronic Equipment.
MIL-DTL-45204	-	Gold Plating, Electrodeposited.
MIL-PRF-22750	-	Coating, Epoxy, High Solids.
MIL-PRF-23586	-	Sealing Compound (With Accelerator) , Silicone Rubber, Electrical.
MIL-H-28719	-	Header, Hermetically Sealed.
MIL-PRF-39012	-	Connectors, Coaxial, Radio Frequency, General Specification for.

(See supplement 1 for list of associated specification sheets).

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment.
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454	-	General Guidelines for Electronic Equipment.

(Copies of these documents are available online at https://assist.daps.dla.mil or from Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) INTERNATIONAL

ASTM-A240/A240M	-	Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications.
ASTM-A484/A484M	-	Steel, Bars, Billets and Forgings, Stainless.
ASTM-A582/A582M	-	Bars, Free-Machining Stainless Steel.
ASTM-A666	-	Steel, Sheet, Strip, Plate, and Flat Bar, Austenitic Stainless, Annealed or Cold-Worked.
ASTM-A693	-	Precipitation-Hardening Stainless and Heat-Resisting Steel Plate, Sheet, and Strip.
ASTM-A967	-	Chemical Passivation Treatments for Stainless Steel Parts.
ASTM-B16/B16M	-	Rod, Brass, Free-Cutting Bars and Shapes for Use in Screw Machines.
ASTM-B36/B36M	-	Plate, Brass, Sheet, Strip, and Rolled Bar.
ASTM-B121/B121M	-	Plate, Leaded Brass, Sheet, Strip, and Rolled Bar.
ASTM-B124/B124M	-	Copper and Copper Alloy Forging Rod, Bar, and Shapes.
ASTM-B209	-	Aluminum and Aluminum-Alloy Sheet and Plate.
ASTM-B221	-	Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles and Tubes.
ASTM-B241/B241M	-	Aluminum and Aluminum-Alloy Seamless Pipe and Seamless Extruded Tube.
ASTM-B339	-	Pig Tin.

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ASTM-B488	-	Gold for Engineering Uses, Electrodeposited Coatings of.
ASTM-B545	-	Tin, Electrodeposited Coatings of.
ASTM-B700	-	Electrodeposited Coatings of Silver For Engineering Use.
ASTM-G21	-	Materials to Fungi, Synthetic Polymeric, Determining Resistance of.

(Copies of these documents are available from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or http://www.astm.org.)

NATIONAL CONFERENCE OF STANDARDS LABORATORIES (NCSL) INTERNATIONAL

NCSL-Z540.3	-	Requirements for the Calibration of Measuring and Test Equipment.

(Copies of this document are available from National Conference of Standards Laboratories (NCSL) International, 1800 30th Street, Suite 305, Boulder, CO 80301-1026; or http://www.ncsli.org/.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

SAE-AMS2404	-	Nickel, Electroless, Plating.
SAE-AMS-I-23011	-	Iron-Nickel Alloys for Sealing to Glasses and Ceramics.
SAE-AMS2422	-	Plating, Gold.
SAE-AMS-QQ-A-200	-	Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube, and Wire, Extruded: General Specification for.
SAE-AMS-QQ-A-250	-	Aluminum and Aluminum Alloy, Plate and Sheet.
SAE-AMS-QQ-N-290	-	Nickel Plating (Electrodeposited) .
SAE-AMS2700		Passivation of Corrosion Resistant Steels.
SAE-AMS-QQ-S-763	-	Steel Bars, Wire, Shapes and Forgings, Corrosion-Resistant.

(Copies of these documents are available from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or http://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern.

3.2 First article. Items furnished under this specification shall be products that have been tested and passed first article inspection (see 4.5 and 6.3).

3.3 Material. The material shall be as specified (see 3.1). When a definite material is not specified, a material shall be used which will enable the items to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Metals. Metals shall be of corrosion resistant type or suitably processed to resist corrosion.

3.3.1.1 Metal sheet. Unless otherwise specified, the material shall be selected from the following:

a. Aluminum alloy sheets shall conform to composition 6061 of SAE-AMS-QQ-A-250 and ASTM-B209.

b. Extruded aluminum alloy shall conform to composition 6063 (refer to ASTM-B221 and ASTM-B241/B241M) or composition 6061 of SAE-AMS-QQ-A-250 and SAE-AMS-QQ-A-200.

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c. Corrosion-resisting steel, sheets, and strips shall conform to ASTM-A240/A240M, ASTM-A666 and ASTM-A693.

d. Corrosion-resisting steel bar, shapes, and forgings shall conform to SAE-AMS-QQ-S-763, ASTM-A484/A484M, and ASTM-A582

e. Copper alloy sheet shall conform to ASTM-B36/B36M, ASTM-B121/B121M, ASTM-B16/B16M, and ASTM-B124/B124M.

3.3.1.2 Iron nickel. Iron nickel alloys shall conform to SAE-AMS-I-23011 and SAE-AMS-QQ-N-290.

3.3.1.3 Gold. Gold shall conform to MIL-DTL-45204, or SAE-AMS2422 and ASTM-B488.

3.3.1.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be in intimate contact.

3.3.2 Plastic. Plastic shall conform to best manufacturing practice.

3.3.3 Rubber. Rubber shall conform to MIL-PRF-23586.

3.3.4 Fungus inert. Materials used in the construction of power dividers, power combiners, and power divider/combiners herein shall be fungus inert in accordance with ASTM-G21. Fungus inert material is defined as a material which, in all modified states and grades, is not a nutrient to fungi.

3.4 Design and construction. Items furnished under this specification shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Impedance. The nominal impedance shall be as specified (see 3.1).

3.4.2 Operating frequency range. The frequency range shall be as specified (see 3.1).

3.4.3 Ambient temperature. The ambient temperature shall be as specified (see 3.1).

3.4.4 Headers. When headers are required (see 3.1), they shall conform to MIL-H-28719.

3.4.5 Terminals and pins. External terminations shall be aligned and identified as specified (see 3.1).

3.4.6 Coaxial connectors. The material design and construction of coaxial connectors, when required (see 3.1), shall conform to MIL-PRF-39012.

3.4.7 Waveguide flanges. Unless otherwise specified (see 3.1), flanges shall mate electrically and mechanically with flanges covered by MIL-DTL-3922.

3.4.8 Connector caps. All waveguide and coaxial connections shall be sealed with push-on caps to prevent damage and the entrance of moisture or foreign material during storage.

3.4.9 Waveguide. The material design and construction of waveguides used in the items, when required (see 3.1), shall conform to MIL-DTL-85.

3.4.10 Finish.

3.4.10.1 Mating surfaces. Unless otherwise specified (see 3.1), the finish of mating surfaces shall be of gold, nickel, silver, or tin in accordance with the appropriate specifications: MIL-DTL-45204, SAE-AMS2422, ASTM-B488, SAE-AMS2404, SAE-AMS-QQ-N-290, ASTM-B700, ASTM-B545 (as restricted, see 3.4.10.3) or ASTM-B339 (see 3.4.10.3). Nickel shall be used only if other materials are unable to meet performance requirements of this specification.

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3.4.10.2 External surfaces. If an external surface is painted, it shall be painted with semigloss or dull black enamel in accordance with type II of MIL-DTL-14072, or primed in accordance with MIL-PRF-22750 and coated in accordance with best manufacturing practice (see 3.1). Aluminum alloy surfaces shall be chemically treated in accordance with MIL-DTL-5541, anodic coatings or passivation treatments in accordance with MIL-A-8625 and ASTM-A967 or SAE-AMS2700, respectively.

3.4.10.3 Pure tin. The use of pure tin, as an underplate or final finish, is prohibited both internally and externally. Tin content of power divider, power combiner and power divider/combiner components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.7).

3.4.11 Threaded parts. All threaded parts shall have screw threads in the unified screw thread series in accordance with FED-STD-H28.

3.4.12 Weight. The weight, when required, shall be as specified (see 3.1).

3.4.13 Life. The manufacturer shall submit to the acquiring activity evidence that items are designed to operate for a period of 5 years and that electrical performance characteristics degradation for this period of 5 years shall not exceed the specified limits (absolute maximum or minimum ratings, see 3.1). This evidence shall be in the form of a certificate to the acquiring activity on each item for which first article is requested and a written guarantee to the acquiring activity on each awarded contract. Any acquired item that fails to meet this life requirement shall be replaced by the manufacturer at no cost to the Government.

3.5 Electrical performance.

3.5.1 Average coupling. When the items are tested as specified in 4.7.3, the arithmetic average of the measured coupling (dB) to the outputs shall be as specified (see 3.1) at any frequency over the specified frequency range.

3.5.2 Voltage standing wave ratio (VSWR). When the items are tested as specified in 4.7.4, the VSWR of each port shall be not greater than the value specified (see 3.1).

3.5.3 Insertion loss. When the items are tested as specified in 4.7.5, the insertion loss (dB) shall not exceed the value specified (see 3.1).

3.5.4 Isolation. When the items are tested as specified in 4.7.6.1, 4.7.6.2, or 4.7.6.3, the isolation (dB) shall be not less than the value specified (see 3.1).

3.5.5 Phase balance. When the items are tested as specified in 4.7.7.1, 4.7.7.2, or 4.7.7.3, the phase balance (°) shall be as specified (see 3.1).

3.5.6 Amplitude balance. When the items are tested as specified in 4.7.8.1 or 4.7.8.2, the amplitude balance (dB) shall be as specified (see 3.1).

3.5.7 Power level. When the items are tested as specified in 4.7.9, there shall be no evidence of breakdown, charring, or arcing. Following this test, VSWR, insertion loss, and isolation values shall be as specified (see 3.1).

3.5.8 Electromagnetic interference (when specified, see 3.1). When the items are tested as specified in 4.7.10, the electromagnetic interference shall be within the limits of MIL-STD-461.

3.6 Environmental.

3.6.1 Seal. The sealing method shall be as specified (see 3.1).

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3.6.1.1 Hermetic seal (when specified, see 3.1). When the items are tested as specified in 4.7.2.1, the leakage rate shall not exceed the following:

Sealed volume case	Maximum allowable leakage
Greater than 2 cubic inches	10^{-6} atm cm^3/s
2 cubic inches or less	10^{-8} atm cm^3/s

The volume shall be computed using the external dimensions of the case, disregarding any mounting (screws, stud, etc.). Following this test, the VSWR, insertion loss, and isolation values shall be as specified in 3.5.2, 3.5.3, and 3.5.4, respectively.

3.6.1.2 O-ring, solder seal, or encapsulated or enclosed seal (when specified, see 3.1). When the items are tested as specified in 4.7.2.2, the VSWR, insertion loss, and isolation values shall be as specified in 3.5.2, 3.5.3, and 3.5.4, respectively.

3.6.2 Barometric pressure (unless otherwise specified, see 3.1). When the items are tested as specified in 4.7.11, there shall be no evidence of breakdown, charring, or arcing. Following this test, the VSWR, insertion loss, isolation, and phase, amplitude balance values shall be as specified in 3.5.2, 3.5.3, 3.5.4, 3.5.5, and 3.5.6, respectively.

3.6.3 Thermal shock (unless otherwise specified, see 3.1). When the items are tested as specified in 4.7.12, the VSWR, insertion loss, and isolation values shall be as specified in 3.5.2, 3.5.3, and 3.5.4, respectively.

3.6.4 Terminal strength. When terminals are tested in accordance with 4.7.13, there shall be no mechanical damage or evidence of loosening or rupturing. Bends shall not be considered as damage unless surface cracking is evident.

3.6.5 Resistance to solvents. When the items are tested in accordance with 4.7.14, there shall be no evidence of mechanical damage and the marking shall remain legible.

3.6.6 Resistance to soldering heat. When the items are tested in accordance with 4.7.15, there shall be no evidence of mechanical damage and the VSWR, insertion loss, and phase and amplitude balance values shall be as specified in 3.5.2, 3.5.3, 3.5.4, 3.5.5, and 3.5.6, respectively.

3.6.7 Solderability. When terminations are tested for solderability as specified in 4.7.16, they shall meet the evaluation criteria of the cited test method.

3.6.8 Vibration. When items are tested as specified in 4.7.17, there shall be no evidence of physical damage and the VSWR, insertion loss, isolation, and phase and amplitude balance values shall be as specified in 3.5.2, 3.5.3, 3.5.4, 3.5.5, and 3.5.6, respectively.

3.6.9 Shock. When items are tested as specified in 4.7.18, the VSWR, insertion loss, isolation, phase and amplitude balance values shall be as specified in 3.5.2, 3.5.3, 3.5.4, 3.5.5, and 3.5.6, respectively.

3.6.10 Moisture resistance. When items are tested as specified in 4.7.19, the phase and amplitude balance, and power level values shall be as specified in 3.5.5, 3.5.6, and 3.5.7, respectively.

3.6.11 Salt atmosphere (unless otherwise specified, see 3.1). This test may be performed on empty flatpacks or pin packages. When the items are tested as specified in 4.7.20, there shall be no evidence of destructive corrosion.

3.6.12 Sand and dust. When the items are tested as specified in 4.7.21, there shall be no sand and dust accumulation within the enclosure and the phase and amplitude balance, and power level values shall be as specified in 3.5.5, 3.5.6, and 3.5.7, respectively.

3.6.13 Explosion (when applicable, see 3.1). When the items are tested as specified in 4.7.22, there shall be no ignition of the ambient atmosphere.

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3.6.14 Temperature cycling screen. When items are tested as specified in 4.7.23, VSWR (4.7.4), insertion loss (4.7.5), and isolation (4.7.6) tests shall be conducted following the temperature cycling screen, to confirm success or failure.

3.7 Marking. The items shall be marked in accordance with MIL-STD-1285, method I, with the part or identifying number (PIN) and the manufacturer's source code (CAGE). Marking characters shall be approximately .125 inch (0.32 mm) in height. The marking shall be placed on the identification plate, using a method which will provide legible and permanent marking for the life of the item. The manufacturer's name or trademark may also be included in the marking provided such is not expressly forbidden in the contract. When space does not permit use of an identification plate or when component usage (eg. space applications) forbids use of a label or plate, marking may be applied directly on a flat or cylindrical surface of the body. Letter size may be reduced to accommodate the following:

PREFERRED	M23971/	military designator and specification number
PIN	XX-YY	specification sheet (XX) and dash (YY) no.
	ZZZZZ	Manufacturer’s source code (CAGE)
PERMISSIBLE	M	military designator
PIN	23971/	specification number
	XX-YY	specification sheet (XX) and dash (YY) no.
	ZZZZZ	Manufacturer’s source code (CAGE)

3.7.1 Special marking. When acquisition requirements (see 6.2) specify, items shall be specially marked with a four digit date code denoting the last two digits of the calendar year (first two digits) and the week of the calendar year (last two digits) when the item was manufactured. Date code marking facilitates process improvement, failure analysis and reliability enhancement.

3.8 Serialization. When the contract requires the items be serialized, each item shall be marked with a unique serial number assigned consecutively within the inspection lot, allowing traceability to the item and test results.

3.9 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.10 Workmanship. The items shall be manufactured and processed in such a manner as to be uniform in quality, and external surfaces shall be free from tool marks, burrs, deep scratches, and other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with NCSL-Z540.3.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspections (see 4.3).

b. First article inspection (see 4.5).

c. Conformance inspection (see 4.6).

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4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table I, used in fabricating the items are in accordance with applicable referenced specifications or requirements prior to such fabrication.

TABLE I. Materials inspection.

Material	Requirement paragraph	Applicable specification
Aluminum	3.3.1.1	ASTM-B221, ASTM-B241/B241M, SAE-AMS-QQ-A-200, SAE-AMS-QQ-A-250, ASTM-B209
Copper alloy	3.3.1.1	ASTM-B121/B121M, ASTM-B36/B36M, ASTM-B16/B16M, ASTM-B124/B124M
Steel bar, shapes, and forgings	3.3.1.1	SAE-AMS-QQ-S-763, ASTM-A484/A484M, ASTM-A582/A582M
Steel sheets and strips	3.3.1.1	ASTM-A240/A240M, ASTM-A666, ASTM-A693
Iron nickel alloy (KOVAR)	3.3.1.2	SAE-AMS-I-23011, SAE-AMS-QQ-N-290
Gold	3.3.1.3	MIL-DTL-45204, SAE-AMS2422, ASTM-B488
Plastics	3.3.2	Best manufacturing practice
Rubber	3.3.3	MIL-PRF-23586

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.5 First article inspection. First article inspection shall be performed by the contractor, after award of the contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted unless extended by the Government to other contracts (see 6.3).

4.5.1 Sample size. Four samples shall be subjected to first article inspection.

4.5.2 Inspection routine. The samples shall be subjected to the inspections specified in table II in the order shown. All sample units shall be subjected to the inspection of group I. The sample shall be divided into 2 groups of 2 units each as specified in table II for groups II and III.

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I (all samples)		
Visual and mechanical	3.1, 3.3-3.4.9, 3.7, 3.10	4.7.1
Seal	3.6.1	4.7.2
Average coupling	3.5.1	4.7.3
VSWR	3.5.2	4.7.4
Insertion loss	3.5.3	4.7.5
Isolation	3.5.4	4.7.6
Phase balance	3.5.5	4.7.7
Amplitude balance	3.5.6	4.7.8
Group II (2 samples)		
Power level	3.5.7	4.7.9
Electromagnetic interference 1/	3.5.8	4.7.10
Barometric pressure	3.6.2	4.7.11
Thermal shock	3.6.3	4.7.12
Terminal strength 5/	3.6.4	4.7.13
Resistance to solvents	3.6.5	4.7.14
Group III (2 samples)		
Solder heat 6/	3.6.6	4.7.15
Solderability 6/	3.6.7	4.7.16
Vibration	3.6.8	4.7.17
Shock	3.6.9	4.7.18
Moisture resistance 2/	3.6.10	4.7.19
Salt atmosphere 4/	3.6.11	4.7.20
Sand and dust 3/	3.6.12	4.7.21
Explosion 4/	3.6.13	4.7.22

1/ When specified.
2/ Applicable to enclosed type and immersion proof only.
3/ Applicable to enclosed type only.
4/ Applicable to hermetic seal and immersion proof only.
5/ Applicable to plug-in and solder terminals.
6/ Applicable to items with solder terminals.

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4.5.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

4.5.4 Disposition of first article sample units. Sample units which have been subjected to first article testing shall not be delivered on any contract or purchase order. The Government reserves the right to retain the sample units or to require the supplier to furnish the sample units with the first article test report.

4.6 Conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.6.1.1 Inspection lot. An inspection lot shall consist of all items covered by a single specification sheet produced under essentially the same conditions, and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table III, in the order shown.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test method Paragraph
Visual and mechanical - - - - - - -	3.1, 3.3, 3.4, 3.7, 3.10	4.7.1
Temperature cycling screen- - - -	3.6.14	4.7.23
Average coupling - - - - - - - - - - -	3.5.1	4.7.3
VSWR - - - - - - - - - - - - - - - - - - -	3.5.2	4.7.4
Insertion loss - - - - - - - - - - - - -	3.5.3	4.7.5
Isolation- - - - - - - - - - - - - - - - -	3.5.4	4.7.6
Phase balance - - - - - - - - - - - - -	3.5.5	4.7.7
Amplitude balance - - - - - - - - - -	3.5.6	4.7.8
Seal - - - - - - - - - - - - - - - - - - -	3.6.1	4.7.2

4.6.1.2.1 Group A Sampling plan. Statistical sampling and inspection shall be performed on an inspection lot basis with a random sample of components selected in accordance with table IV. The acceptance levels shall be based upon the zero defective sampling plan. No failures shall be permitted. Sample size is for Standard Military component reliability components. Components intended for “Space Flight Applications” are defined as those with the highest reliability level provided by this specification (sample size = 100 percent).

TABLE IV. Group A sampling plan.

Lot size	Sample size (Standard Military)	“Space Flight Application” power divider Sample size (highest reliability level)
1 - 13	100 percent	100 percent
14 - 150	13	100 percent
151-280	20	100 percent
281-500	29	100 percent
501-1,200	34	100 percent
1,201-3,200	42	100 percent
3,201-10,000	50	100 percent

4.6.1.2.2 Group A Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table V, in the order shown, and the sample shall be selected from inspection lots that have passed group A inspection.

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TABLE V. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Power level	3.5.7	4.7.9
VSWR	3.5.2	4.7.4
Insertion loss	3.5.3	4.7.5
Isolation	3.5.4	4.7.6
Electromagnetic interference 1/	3.5.8	4.7.10
Thermal shock	3.6.3	4.7.12
VSWR	3.5.2	4.7.4
Insertion loss	3.5.3	4.7.5
Isolation	3.5.4	4.7.6
Terminal strength 4/	3.6.4	4.7.13
Resistance to solvents	3.6.5	4.7.14
Solderability 5/	3.6.7	4.7.16
Vibration	3.6.8	4.7.17
VSWR	3.5.2	4.7.4
Insertion loss	3.5.3	4.7.5
Isolation	3.5.4	4.7.6
Phase balance	3.5.5	4.7.7
Amplitude balance	3.5.6	4.7.8
Shock	3.6.9	4.7.18
VSWR	3.5.2	4.7.4
Insertion loss	3.5.3	4.7.5
Isolation	3.5.4	4.7.6
Phase balance	3.5.5	4.7.7
Amplitude balance	3.5.6	4.7.8
Moisture resistance 2/	3.6.10	4.7.19
VSWR	3.5.2	4.7.4
Insertion loss	3.5.3	4.7.5
Isolation	3.5.4	4.7.6
Phase balance	3.5.5	4.7.7
Amplitude balance	3.5.6	4.7.8
Salt atmosphere 3/	3.6.11	4.7.20

1/ When specified.
2/ Applicable to enclosed type and immersion proof only.
3/ Applicable to hermetic seal and immersion proof only.
4/ Applicable to plug-in and solder terminals.
5/ Applicable to items with solder terminals.

4.6.1.3.1 Group B Sampling plan. Statistical sampling and inspection shall be performed on an inspection lot basis with a random sample of components selected in accordance with table VI. Acceptance shall be based upon the zero defective sampling plan. No failures shall be permitted. Sample sizes are for standard Military component reliability and for higher level reliability (intended for space flight applications).

TABLE VI. Group B sampling plan.

Lot size	Sample size (Standard Military)	“Space Flight Application” power divider sample size
2-25	1	2
26-150	2	4
151-10,000	4	8

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4.6.1.3.2 Group B Rejected items. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.3.3 Disposition of sample units. Sample items which have been subjected to group B inspection shall not be delivered on the contract.

4.6.1.3.4 Noncompliance. If a sample fails to pass group B inspection, the manufacturer shall notify the acquiring activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of product shall be discontinued until corrective action, acceptable to the acquiring activity, has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the acquiring activity. Group A inspection may be reinstated; however, final acceptance and shipment shall be withheld until the group B re-inspection has shown that the corrective action was successful. In the event of failure after re-inspection, information concerning the failure and the corrective action taken shall be furnished to the cognizant acquiring activity (see 6.4).

4.7 Methods of inspection.

4.7.1 Visual and mechanical inspection. The items shall be examined to verify that the material, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.7, and 3.10).

4.7.2 Seal.

4.7.2.1 Hermetic seal (see 3.6.1.1). Hermetically sealed items shall be tested in accordance with MIL-STD-202, method 112. The following apply:

a. Test conditions letter: C.

b. Procedure: III.

c. Degree of leakage rate sensitivity (see 3.6.1.1).

4.7.2.2 O-ring, solder seal, or encapsulated or enclosed seal (see 3.6.1.2). O-ring seal, solder seal, or encapsulated sealed items shall be tested in accordance with MIL-STD-202, method 104. The following applies:

a. Test condition letter: B.

4.7.3 Average coupling (see 3.5.1). The average coupling of the item shall be determined by the sweep frequency technique over the specified frequency range (see 3.1) using the test setup of figure 1 or equivalent. The nominal coupling loss for each type of toroidal, stripline and micro-strip power divider shall be as follows:

Table VII. Nominal coupling loss values for toroidal, stripline and micro-strip power dividers.

Type of power divider	Nominal coupling loss (dB)
2-Way	3.01
Resistive Loaded 2-Way	6.0
3-Way	4.77
4-Way	6.02
6-Way	7.78
8-Way	9.03
12-Way	10.80
16-Way	12.04
24-Way	13.80

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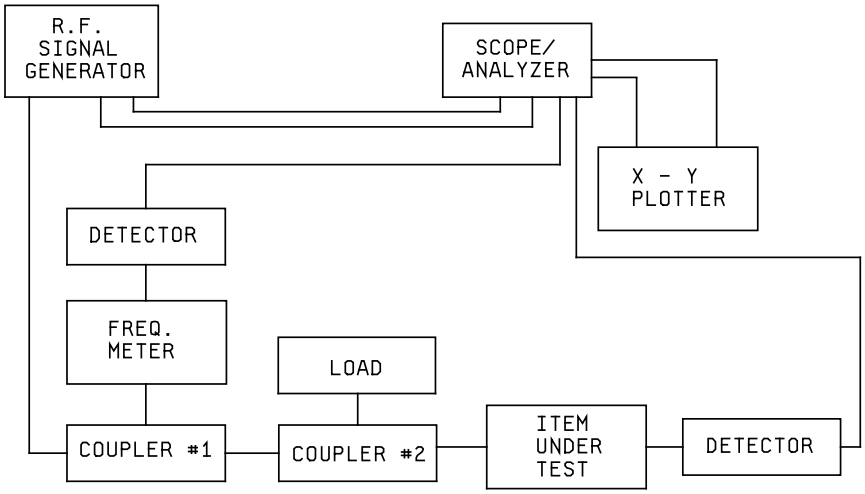


FIGURE 1. Sweep frequency coupling test setup (for reference only).

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4.7.4 VSWR (see 3.5.2). The VSWR of each port shall be measured across the full frequency range (see 3.1) using sweep frequency technique and the test setup of figure 2 or equivalent. Each unused port shall be terminated in a matched load. VSWR may be measured at fixed frequencies using figure 3 or equivalent, at seven equally spaced points across the frequency range (see 3.1). The accuracy of each measurement shall be within .1 dB.

4.7.5 Insertion loss (see 3.5.3). The insertion loss shall be determined by subtracting the nominal value of the divider from the measured average coupling value (see 4.7.3).

4.7.6 Isolation (see 3.5.4). The isolation (in dB) shall be measured using the sweep frequency technique across the full frequency range (see 3.1) of the item, using the test setup of figure 4 or equivalent. The isolation may be measured at 7 equally spaced points across the frequency range of the items (see 3.1). The accuracy of each measurement shall be within .1 dB. The procedures for measuring the isolation of the various items are as follows.

4.7.6.1 Power divider, quadrature. The isolation shall be measured between any selected port and adjacent isolated port with the opposite output ports terminated with matched loads (VSWR of 1.02). Repeat the measurement for each remaining port.

4.7.6.2 Power combiner. The isolation shall be measured between two input ports with the output port and any additional input ports (if any) terminated with matched loads (VSWR of 1.02). Repeat the measurement for each additional input port.

4.7.6.3 Power divider-combiner. The isolation shall be measured between the symmetric and anti-symmetric ports with the output ports terminated with loads matched to each other (VSWR of 1.02) or in open circuits if matched loads are not obtainable. Next, the isolation shall be measured between two output ports with the remaining output ports, if any, and symmetric and anti-symmetric ports terminated in matched loads (VSWR of 1.02).

4.7.7 Phase balance (see 3.5.5). The phase balance shall be measured at the low end, middle, and high end of the frequency range (see 3.1) using the test setup of figure 5 or equivalent. Suggested test procedures for the various items are as follows.

4.7.7.1 Power divider, quadrature. The phase balance of the item shall be measured between the output ports with the other co-linear output port and the adjacent isolated port terminated in matched loads (VSWR of 1.02). Repeat the measurement for any remaining ports.

4.7.7.2 Power combiner. The phase balance of the item shall be measured as the difference at the output between in-phase signals at two input ports with the remaining input ports terminated in matched loads (VSWR of 1.02). Repeat the measurement for each remaining input port.

4.7.7.3 Power divider-combiner. The phase balance of the item shall be measured between the symmetric port and an output port with the anti-symmetric port and remaining output ports terminated in matched loads (VSWR of 1.02). Repeat the measurement for each remaining output port. Next, measure the phase balance between the anti-symmetric port and an output with the remaining output ports terminated in matched loads (VSWR of 1.02). Repeat the measurements for each output port.

4.7.8 Amplitude balance (see 3.5.6).

4.7.8.1 Power divider, quadrature. Use the test procedure of 4.7.7 and 4.7.7.1 except read the amplitude balance.

4.7.8.2 Power combiner. Use the test procedure of 4.7.7 and 4.7.7.2 except read the amplitude balance.

4.7.9 Power level (see 3.5.7). The specified peak and average power shall be applied simultaneously at any frequency within the specified range (see 3.1). When no peak power is specified, apply 1.5 times rated average power. Power shall be maintained for a period of 15 minutes after the item has reached thermal equilibrium. It is considered that thermal equilibrium has been reached when the temperature of the item has not changed by more than 5°C over a period of 5 minutes. For pressurized items, the internal pressure shall be as specified (see 3.1). Following the exposure, the VSWR, insertion loss, and isolation shall then be measured as specified in 4.7.4, 4.7.5, and 4.7.6, respectively.

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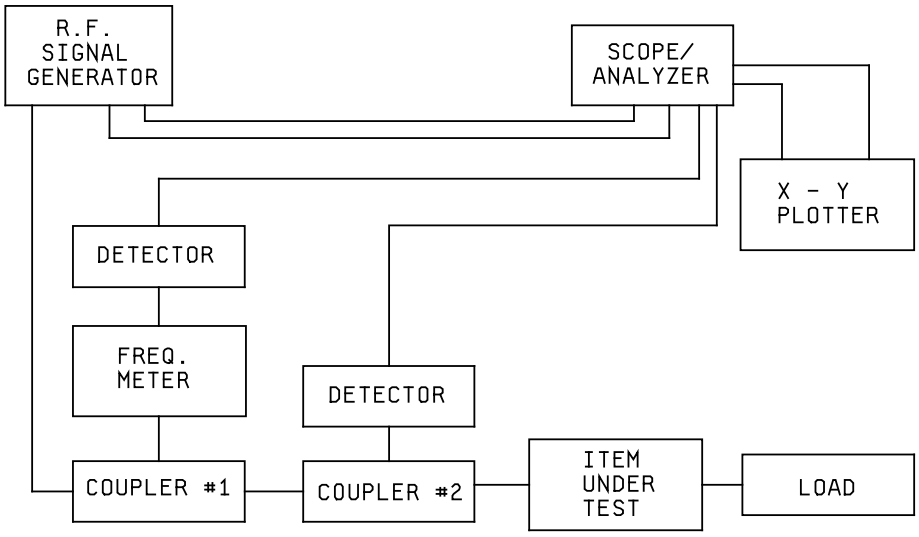


FIGURE 2. Sweep frequency VSWR test setup (for reference only).

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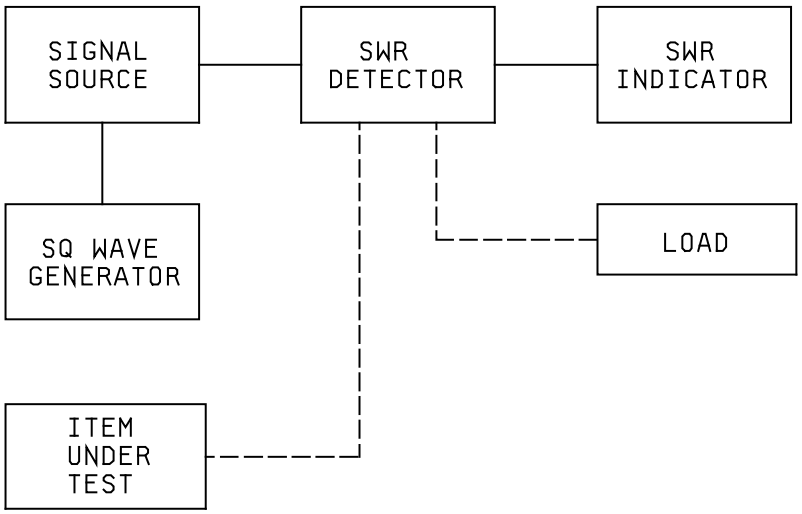


FIGURE 3. VSWR test setup, fixed point method (for reference only).

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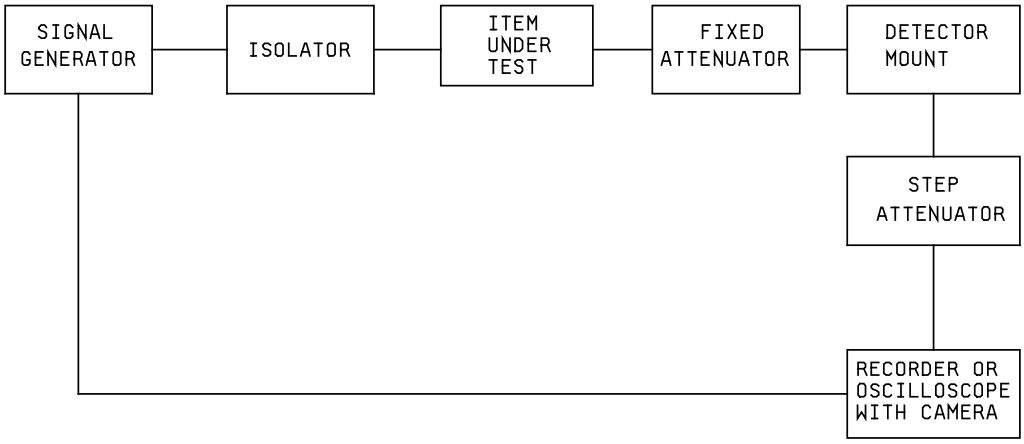


FIGURE 4. Isolation measurement test setup (for reference only).

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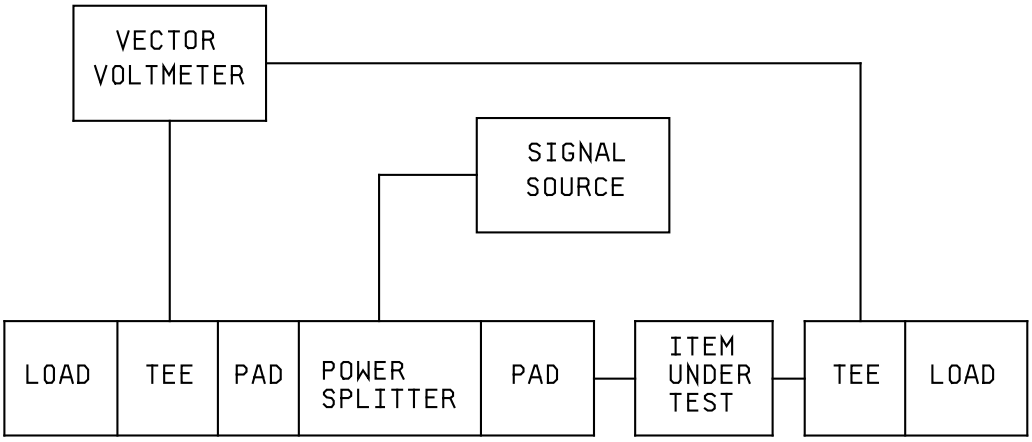


FIGURE 5. Output phase and amplitude balance test setup (for reference only).

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4.7.10 Electromagnetic interference (see 3.5.8). When specified (see 3.1), the electromagnetic interference shall be measured in accordance with CE101, CE102, CS101, and RE101 of MIL-STD-461.

Conducted emission, 30 Hz to 20 kHz, power leads
Conducted emission, 20 kHz to 50 MHz, power leads
Conducted emission, 20 kHz to 50 MHz, control and signal leads
Conducted susceptibility, 30 Hz to 50 kHz, power lead
Conducted susceptibility, 50 kHz to 400 MHz, power lead
Radiated emission, 30 Hz to 30 kHz, magnetic field

4.7.11 Barometric pressure (reduced) (see 3.6.2). The items shall be tested for barometric pressure in accordance with method 105 of MIL-STD-202. The following details shall apply:

a. Mounting: Not applicable.

b. Test condition: D (100,000 feet), unless otherwise specified (see 3.1).

c. Test during subjection to pressure: 125 percent of rated power shall be applied between the terminals for not less than 1 second or more than 5 seconds.

d. Final measurements: VSWR, insertion loss, isolation, phase balance, and amplitude balance tests as specified in 4.7.4, 4.7.5, 4.7.6, 4.7.7, and 4.7.8, respectively.

4.7.12 Thermal shock (see 3.6.3). The items shall be tested in accordance with method 107 of MIL-STD-202, test condition A. Before and after cycling, the VSWR, insertion loss, and isolation tests as specified in 4.7.4, 4.7.5, and 4.7.6, respectively shall be performed.

4.7.13 Terminal strength (see 3.6.4). The items shall be tested in accordance with method 211 of MIL-STD-202, test condition A, with a 2 pound pull.

4.7.14 Resistance to solvents (see 3.6.5). The item shall be tested in accordance with method 215 of MIL-STD-202. Two test solvents may be selected from those defined in method 215, and applied one to each sample. After the final immersion and brushing, it shall be allowed to dry. When dry, there shall be no evidence of mechanical damage and the marking shall remain legible.

4.7.15 Resistance to soldering heat (see 3.6.6). The item shall be tested in accordance with method 210 of MIL-STD-202, test condition D, followed by the VSWR, insertion loss, isolation, phase balance, and amplitude balance tests as specified in 4.7.4, 4.7.5, 4.7.6, 4.7.7, and 4.7.8, respectively.

4.7.16 Solderability (see 3.6.7). The item shall be tested in accordance with method 208 of MIL-STD-202.

4.7.17 Vibration (see 3.6.8). Unless otherwise specified (see 3.1), the item shall be tested in accordance with method 204 of MIL-STD-202, test condition A, followed by the VSWR, insertion loss, phase balance, and amplitude balance tests as specified in 4.7.4, 4.7.5, 4.7.6, 4.7.7, and 4.7.8, respectively.

4.7.18 Shock (see 3.6.9). Unless otherwise specified (see 3.1), the item shall be tested in accordance with method 213 of MIL-STD-202, test condition G, followed by the VSWR, insertion loss, isolation, phase balance, and amplitude balance tests as specified in 4.7.4, 4.7.5, 4.7.6, 4.7.7, and 4.7.8, respectively.

4.7.19 Moisture resistance (see 3.6.10). With all connections capped, the item shall be tested in accordance with method 106 of MIL-STD-202, 1,000 megohms minimum (unless otherwise specified, see 3.1) at +25°C measured after drying followed by the phase balance, amplitude balance, and power level tests as specified in 4.7.7, 4.7.8, and 4.7.9, respectively. Polarization and load voltage is not applicable.

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4.7.20 Salt atmosphere (see 3.6.11). This test may be performed on empty flatpacks or pin packages. The item shall be tested in accordance with method 101 of MIL-STD-202.

a. Apply salt solution: 5 percent.

b. Test condition: B.

c. After test, dry specimens for 24 hours ± .5 at 40°C.

d. Inspect item for corrosion.

4.7.21 Sand and dust (see 3.6.12). The item shall be tested in accordance with method 110 of MIL-STD-202, test condition A. At completion of the exposure period, the device shall be tested for phase balance, amplitude balance, and power level as specified in 4.7.7, 4.7.8, and 4.7.9, respectively.

4.7.22 Explosion (see 3.6.13). The item shall be tested in accordance with method 109 of MIL-STD-202 under rated load (see 3.1).

4.7.23 Temperature cycling screen (see 3.6.14). Temperature cycling screen test shall be conducted using an air atmosphere test chamber, in accordance with MIL-STD-810, and with the following guidelines. Number of temperature cycles shall be ten (10). Each temperature cycle (see figure 6) includes hot and cold temperature soaks, wherein the test item temperature stabilizes within ± 2°C of high and low temperature extremes (maximum and minimum storage temperatures (see 3.1), respectively). Temperature extremes are separated by part cool down and warm up temperature change rate periods of 20°C per minute, chamber permitting. Warm up is included in the "5 hours" interval of the 8 hour cycle. Test items should be non-operating during temperature cycling. VSWR (4.7.4), insertion loss (4.7.5) and isolation (4.7.6) tests should be conducted following the temperature cycling screen to confirm success or failure.

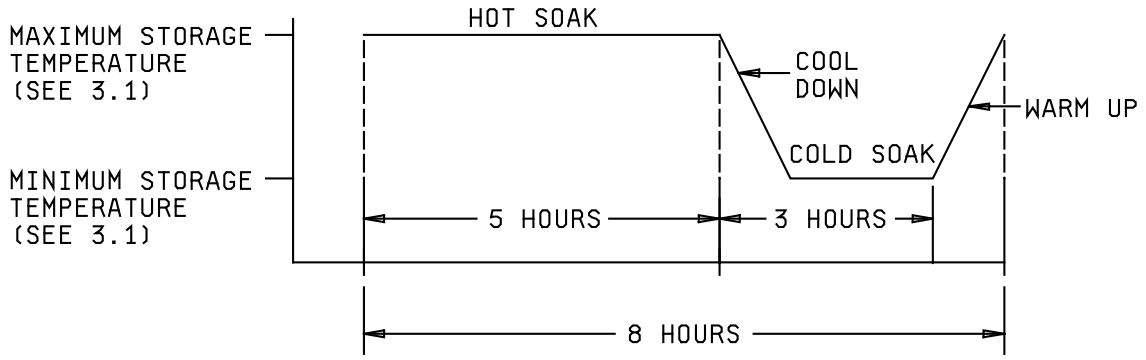


FIGURE 6. Temperature cycling profile.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Items covered by this specification are intended for use in military systems for signal splitters, signal combiners, power adders, homing systems radio direction finding, phase networks, and communication systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. If required, the specific issue of individual documents referenced (see 2.2.1).

c. Packaging requirements (see 5.1).

d. Special marking, if required (see 3.7.1).

e. Level of component reliability (space flight application or standard Military) for Group A and Group B conformance inspection sample size.

6.3 First article. Information pertaining to first article inspection and approval of products covered by this specification should be obtained from the acquiring activity.

6.4 Subject (key word) list:

amplitude
coaxial
communication systems
connectors
corrosion-resisting
coupling
electrical
finish
first article
flanges
frequency range
headers
hermetic
insertion loss
isolation
metals
phase balance
phase networks
power
quadrature
seal
signal splitters
solderability
VSWR
waveguides

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous revision were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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6.6 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmentally Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website at http://www.epa.gov/osw/hazard/wastemin/priority.htm. Included in the list of 31 priority chemicals are cadmium, lead, and mercury. Use of the materials on the list should be minimized or eliminated unless needed to meet the requirements specified herein (see section 3).

6.7 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers (see 3.4.10.3). For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-2012-009)

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